

Dhrumil Patel

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EXPERIENCE

Optiver

Chicago, US

Software Engineer

Feb 2026 - Present

- Designed and deployed a low-latency user application for floor trading
- Built & deployed an agent to custodian a multi-engineer, multi-day corporate action process in **<10 minutes**

Software Engineer Intern

Jun - Aug 2025

- Built and deployed a distributed RPC service serving **>10k requests/s** with **2MB** payloads at low latency
- Investigated production GC incidents and implemented a fix to avoid kernel lock contention

Ramp

May - Aug 2024

Software Engineer Intern

New York City, US

- Built the international bill payment integration for 2 accounting platforms, deployed to **2600+** businesses
- Migrated accounting validation functions to run in bulk, improving their **p95 execution time** by **5x**
- Built a chatbot to search **500k+** leads' websites; won the company hackathon and deployed the project

Snowflake

Sept - Dec 2023

Software Engineer Intern

San Mateo, US

- Built clustering functions for Snowflake SQL to cluster **100M rows x 1k features** with metrics
- Developed a novel algorithm to find the number of clusters in large datasets in **linearithmic** time

Roblox

May - Aug 2023

Software Engineer Intern

San Mateo, US

- Built QA's **first** observability tools to calculate release metrics and report anomalies, used by **4 teams**

Vector Institute

Sept - Dec 2022

Applied Machine Learning Research Intern

Toronto, CA

- Used domain adaptation to improve privacy-preserving federated learning **CNNs'** performance by **3.2%**

A Thinking Ape

May - Aug 2022

Software Engineer Intern

Vancouver, CA

- Implemented **2-step authentication** using **Django**, **Objective C** and **Java** to eliminate **10k+** bots per day

Flinks

Jun 2021 - May 2022

Machine Learning Intern

Montreal, CA

- Built & deployed a **BERT**-driven entity recognition system in **PyTorch** to interpret **100M+** transactions per day
- Improved entity recognition F1-score by **56%** and cut required training data by **30%**

PROJECTS & ORGANIZATIONS

Haystackfs: A verified Discord bot for file search used by **42k+ users** in **900+ servers** May 2021 - Present

Unsupervised Monocular Depth Estimation: Extends [Godard et al. \(2026\)](#) with semantic segmentation 2023

Watolink: Led a team of 10 to build an EEG processing API for mind-controlled communication 2021-2022

Google Code-In: SwagLyrics: Built a **GitHub App** that reported **1500+** unsupported songs Oct 2019 - Feb 2020

AWARDS & ACCOMPLISHMENTS

Top 3% Globally, 7th in Canada at Google Hashcode Qualifying Round 2022

Top 7.5% Globally in Facebook HackerCup 2021

EDUCATION

University of Waterloo, Bachelor of Computer Science, Honours

2025

President's Scholarship of Distinction

TECHNICAL SKILLS

Languages: Python, C++, SQL, Java, Objective C

Tech: BigQuery, Docker, ElasticSearch, Spark, PostgreSQL, PyTorch, OpenCV, scikit-learn, NumPy